

2019(CBCS)

Time : 3 hours

Full Marks : 70

Pass Marks : 32

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

*Answer any **five** questions in which Q. No. 1 is compulsory.*

1. Answer any **four** questions in which **50** words each : $3\frac{1}{2}\times 4 = 14$

   (a) Find Laplace transform of sin $\sqrt{t}$.

   (b) Show that for Hermite polynomial $H_n(x), H'_n(x) = 2nH_{n-1}(x)$.

   (c) Define Fourier's transform.

   (d) Define interpolation and extrapolation.

(e) Give the fundamental terms of Fortran programming.

(f) Define Geodesic.

2. Write down Hermite differential equation and solve it. 14

3. State and prove second shifting theorem in Lapalce transform. Hence, obtain Laplace thransform of delta function. 7+7 = 14

4. Using Lapalce transform solve the given differential equation : 14

$$\frac{d^2y}{dt^2} + 25y = 10 \cos 5t$$

5. Write down Legendre differential equation and solve it. 14

6. Give the Stellar classification and discuss the H. R. diagram of clusters. 14

7. Using the method of least square, find the straight line fits the following data : 14

x	y
1	14
2	27
3	40
4	55
5	68

8. Obtain the formula for Simpson's 1/3 rule and evaluate $\int_0^6 \frac{1}{1+x^2}\,dx$. 7+7 = 14

9. Write short notes on any **two** of the following : 7×2 = 14

   (a) Bessel's functions

   (b) Hubble's law

   (c) Eigen value and eigen function of square matrix

   (d) Flow chart